

BPC Training

Module 1 – Data Analytics & Smart Processing System

9th – 11th Jan 2023
CC-SISPI FTKKP UMP

Personal Background

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PhD (Process Monitoring), Newcastle University, UK

M. Eng. (Hazard Identification), UTM, Malaysia

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Current Position

- Associate Professor, Faculty of Chemical and Process Engineering Technology, UNIVERSITI MALAYSIA PAHANG
- Director, Centre for Sustainability of Ecosystem & Earth Resources (Pusat ALAM), UNIVERSITI MALAYSIA PAHANG



Field of Expertise

No.	Areas of Specialization	Contributions & Achievements
1.	Process Monitoring & Process Chemometrics	Master & degree programs, consultation
2.	Rare Earth Processing & Modelling	PhD and Master program, consultation and technical services
3.	Wastewater Treatment & Electrodialysis System	PhD program and technical services
4.	Solid Waste Management & Recovery	Consultation and technical services

Topics

1. Data Analytics Background

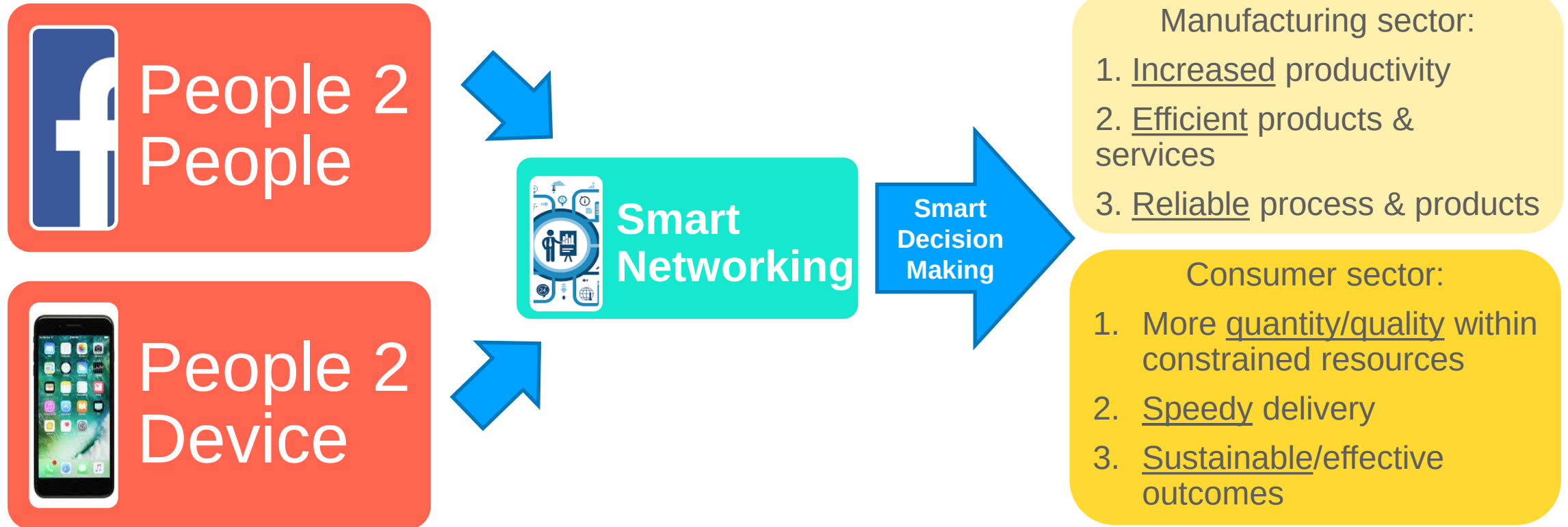
- Data analytics and IR 4.0
- Application of data analytics through smart monitoring application.

2. Smart Processing System

- Feature of smart monitoring application.
- Data reconciliation & pre-treatment

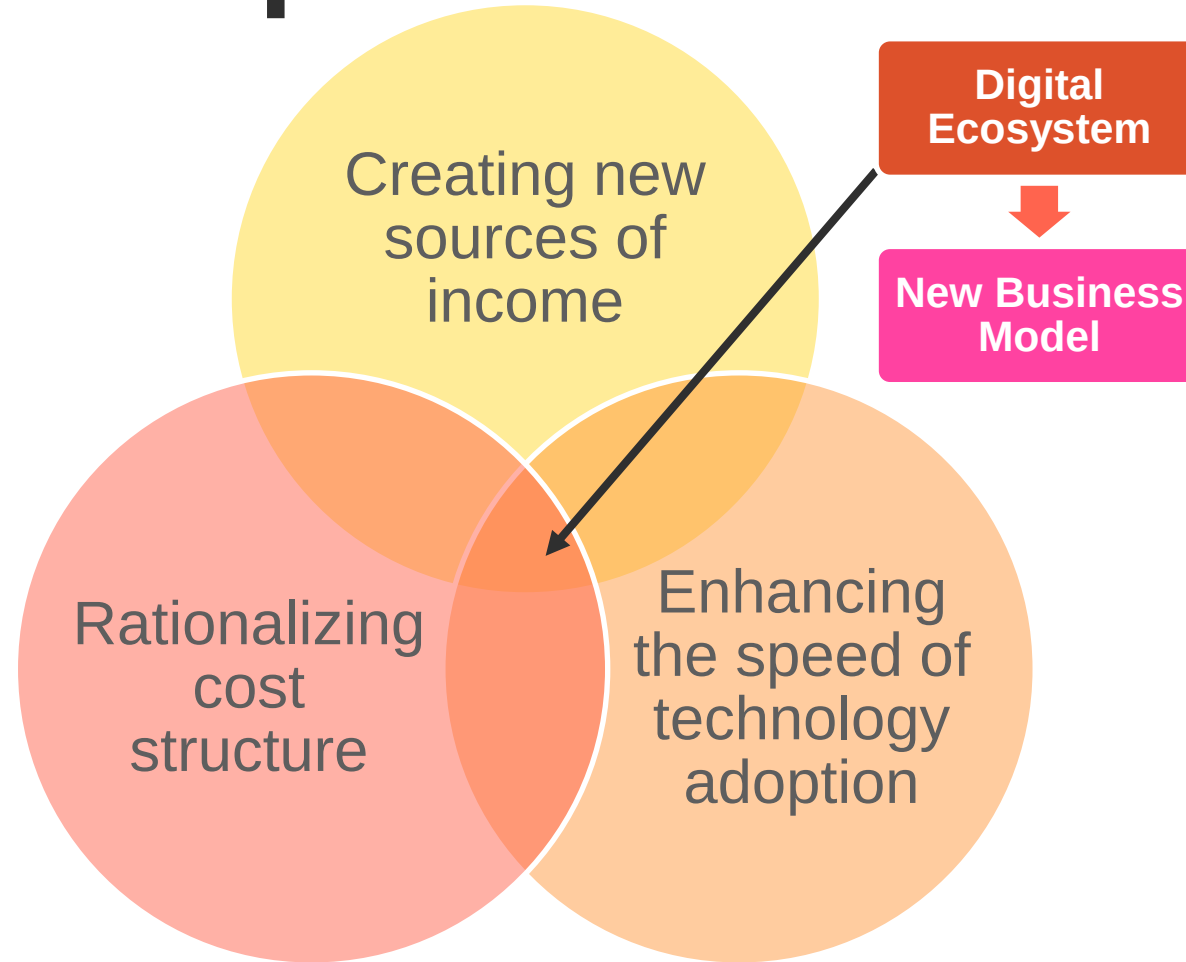
Data Analytics Background

The Radical Challenges of IoT Era



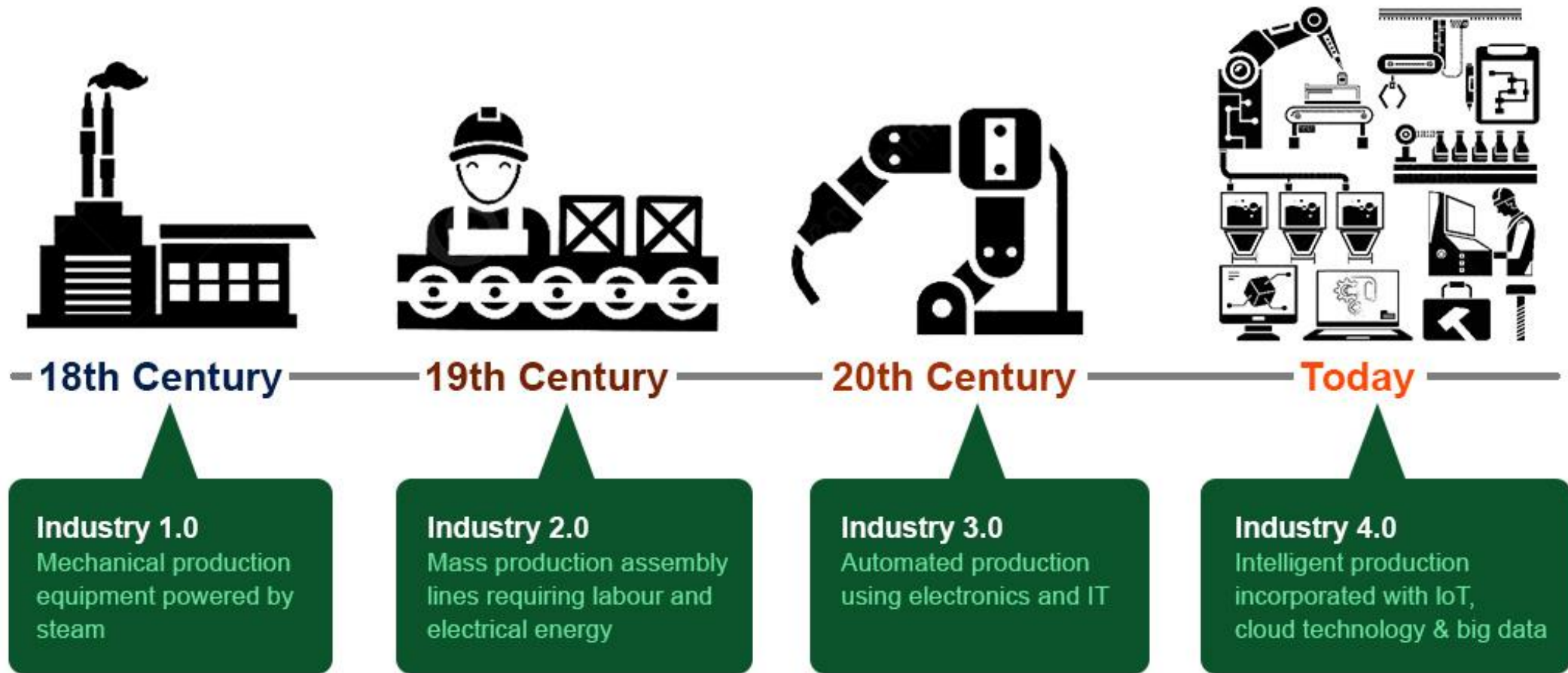
Data Analytics Background

BIG impacts!



Data Analytics Background

Industrial 4.0



<http://www.cans2u.com/industry4.php>

Data Analytics Background

How exactly data analytic will enhance industrial process performance?

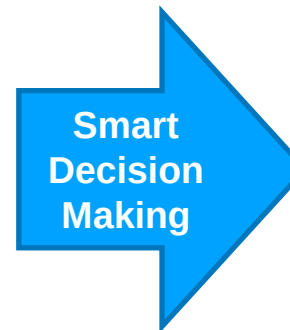


Data Analytics Background

Industrial 4.0 in Smart Manufacturing System



**Multitasking of Data
Driven Approach**

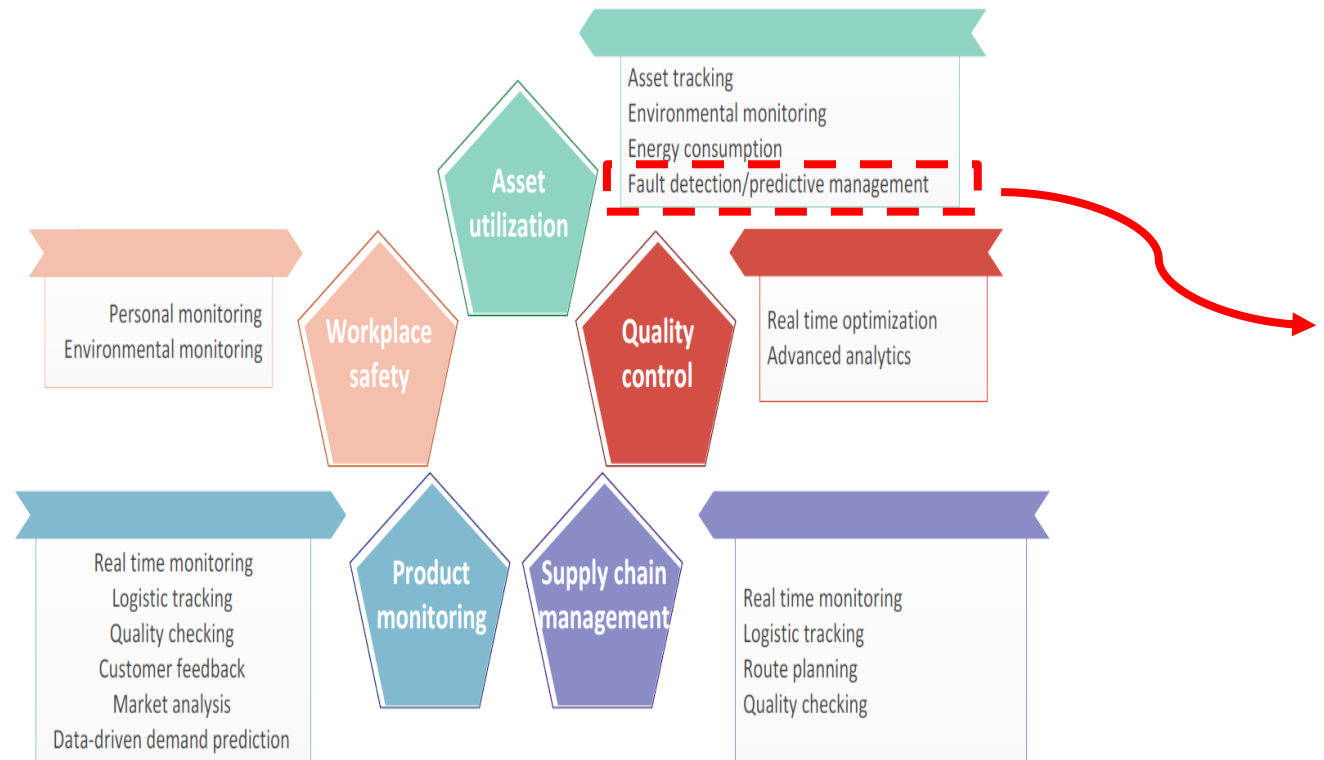


**Decentralized
decisions**

The ability of cyber physical systems to make decisions on their own and to perform their tasks as autonomous as possible.

Data Analytics Background

Application Domains of BIG DATA System

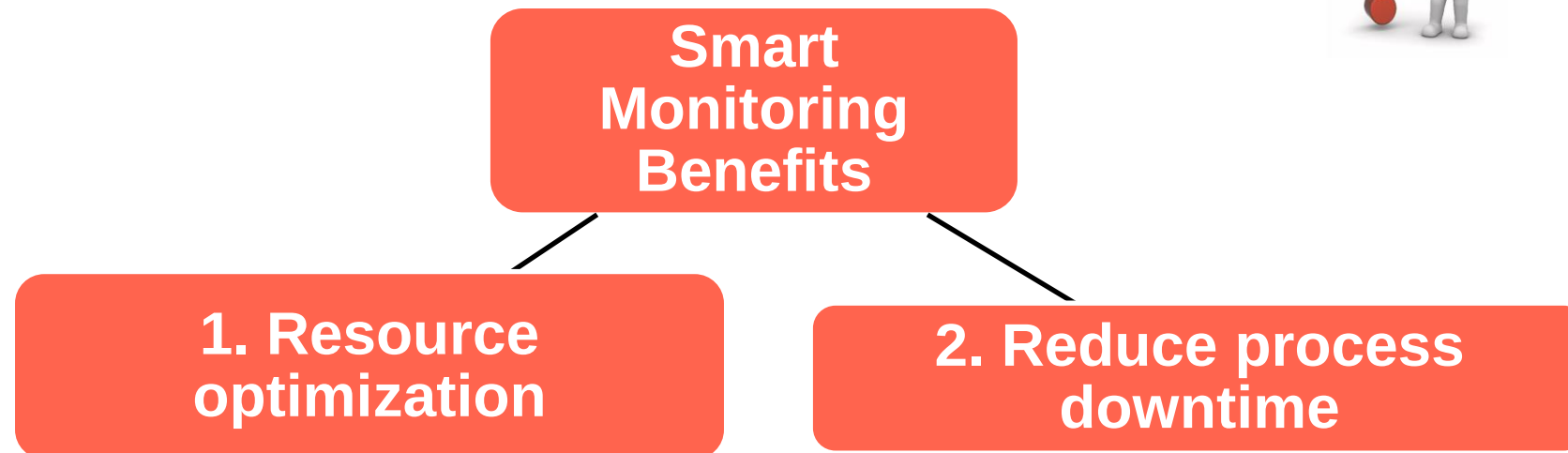


Pilloni, V., (2018). "How Data Will Transform Industrial Processes. Crowdsensing, Crowdsourcing and Big Data as Pillars of Industry 4.0". Future Internet, 10, 24.



SMART MONITORING SYSTEM

What are the direct benefits of smart monitoring system?



Data Analytics Background

1. Resource optimization



Data Source:

Energy consumption parameters & weather forecast information



Connectivity:

Real-time remote service database – correlating internal & external data



Intelligent Mechanism:

Optimal wind turbine operation based on weather forecast

Data Analytics Background

2. Reduce process
downtime



Data Source:
Temperature profile



Connectivity:
Plant data cloud /
imported database



Intelligent Mechanism:
Predictive maintenance
& diagnosis

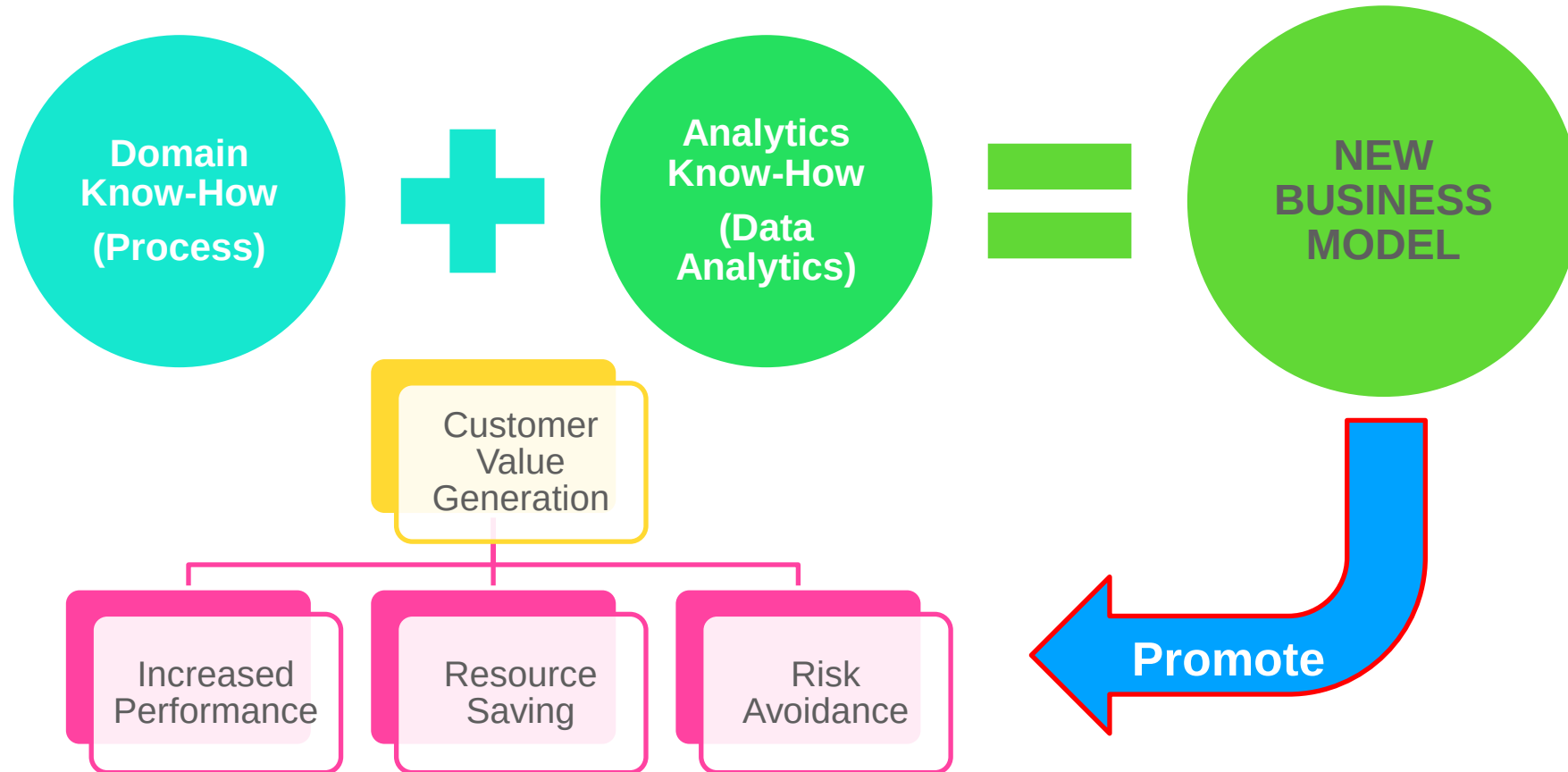
Data Analytics Background

How actually data analytics can drive business performance?



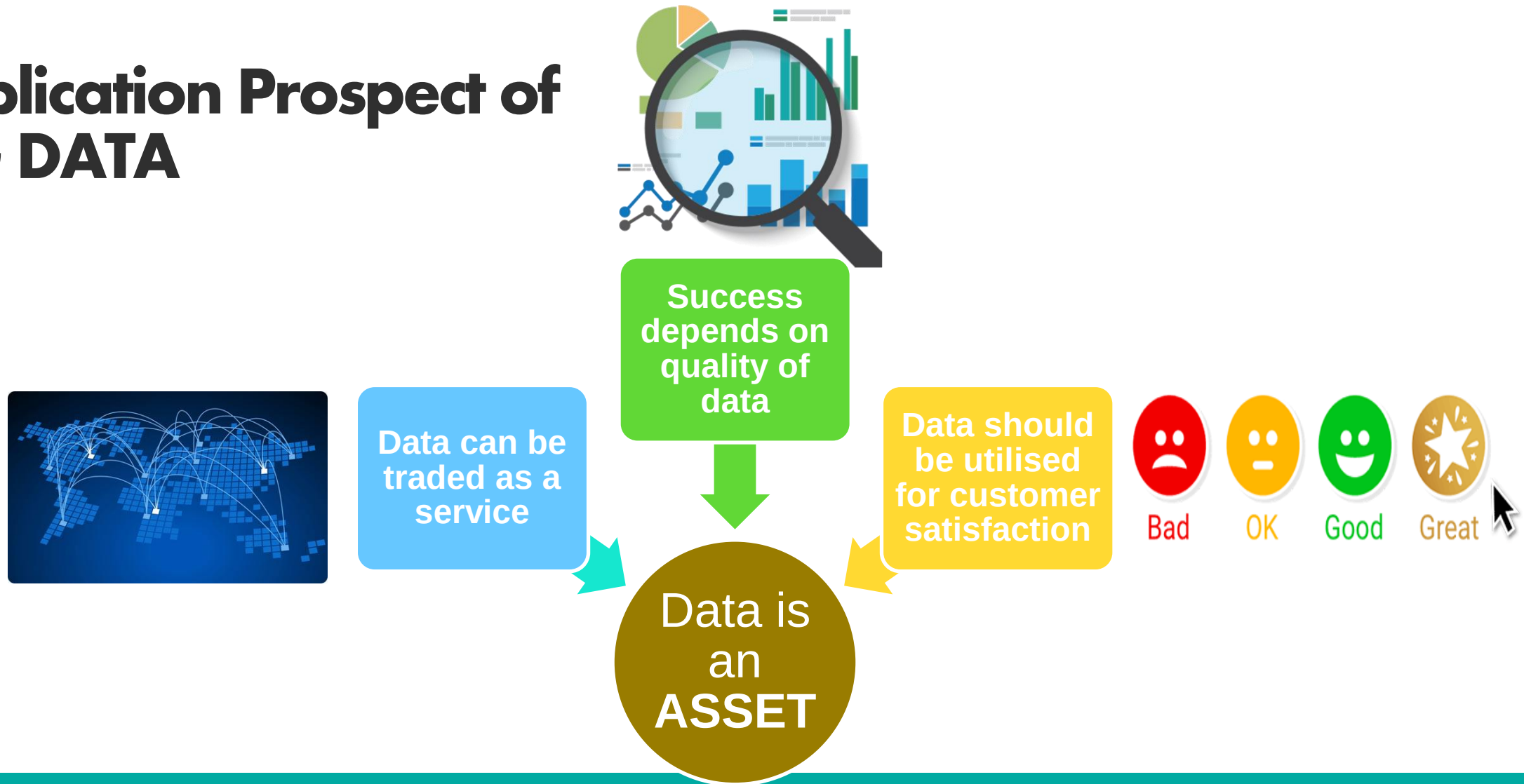
Data Analytics Background

Data Analytics ~> Smart Application ~> Business Analytics



Data Analytics Background

Application Prospect of BIG DATA

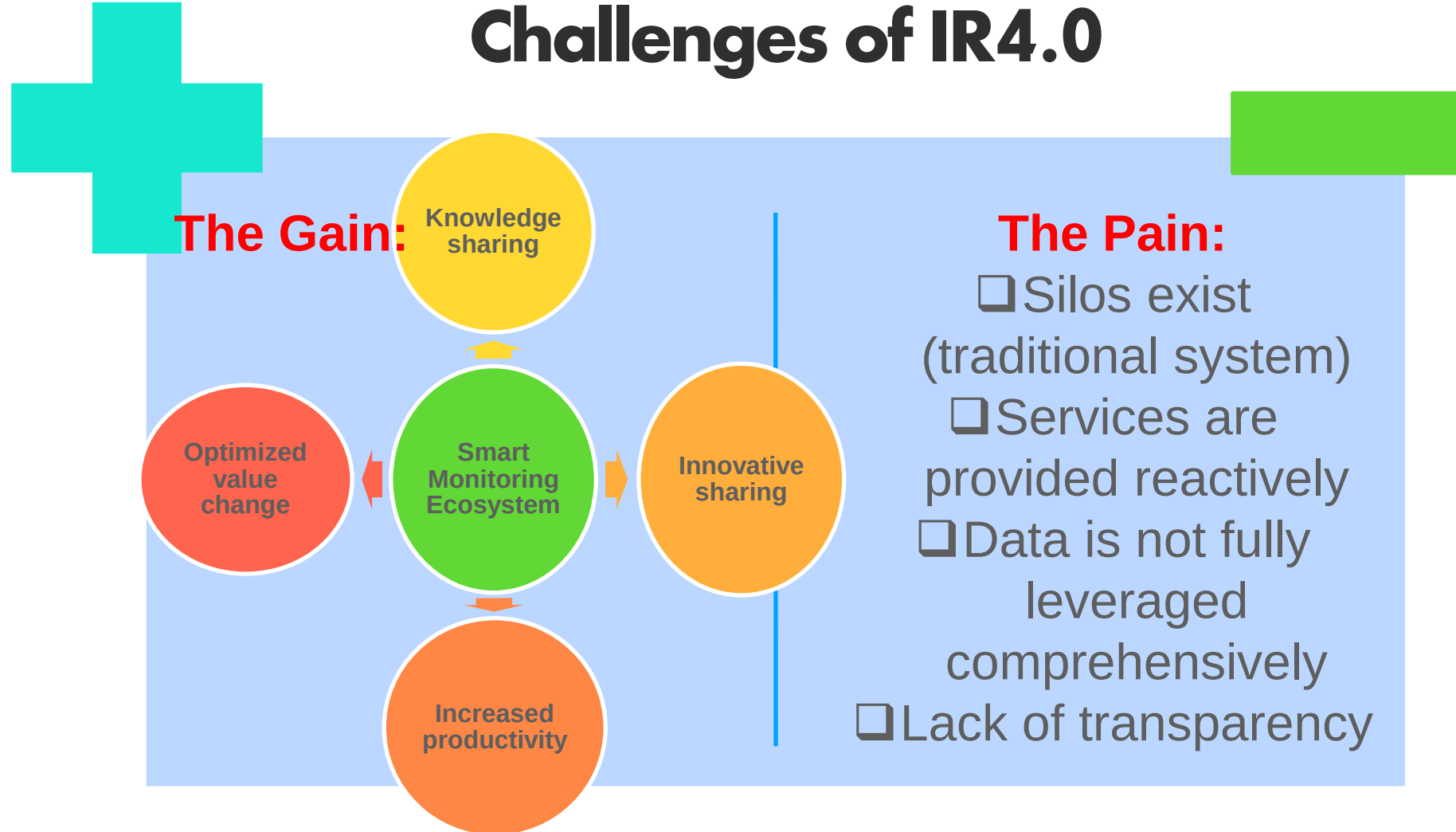


Exercise 1

How does your current operation performance dashboard system utilizes the data analytics functionality?

Smart Processing System

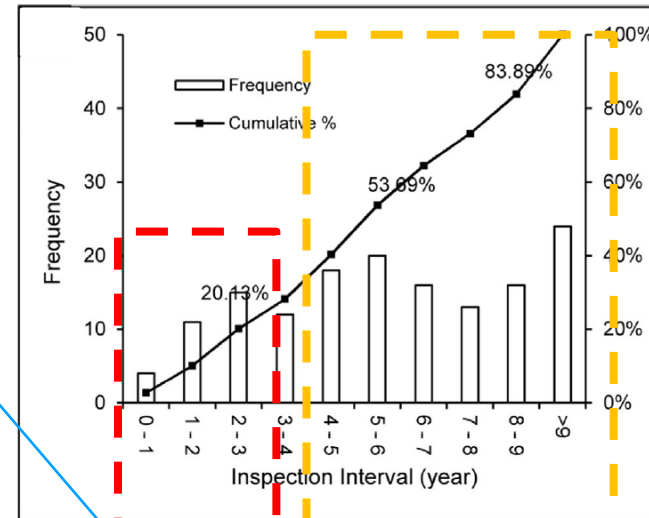
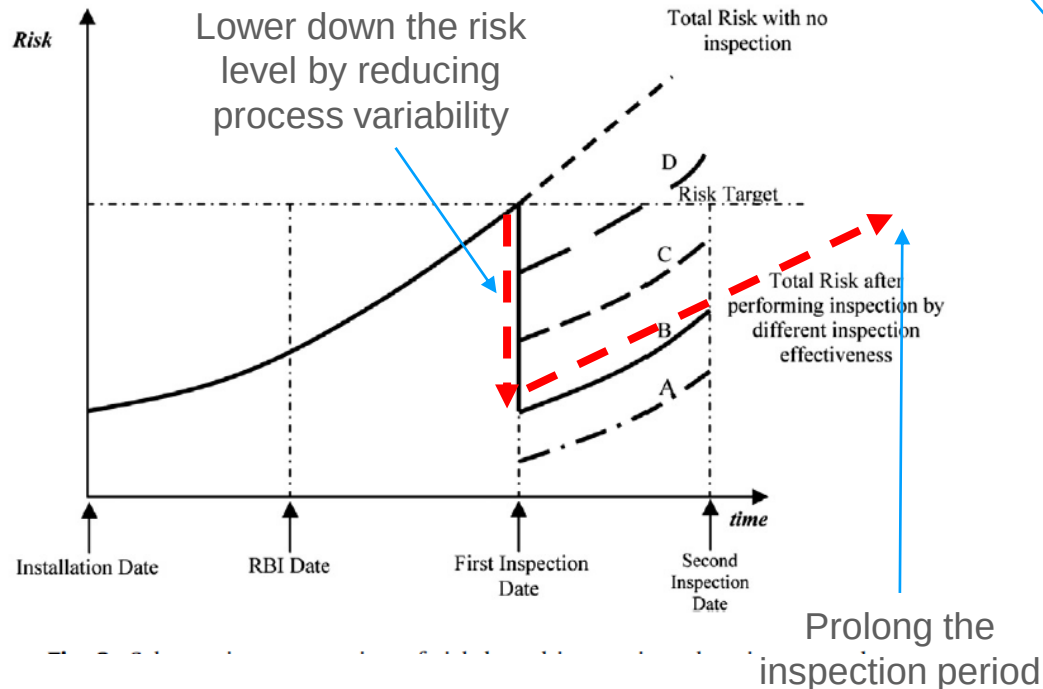
Challenges of IR4.0



Smart Processing System

Feature No. 1: Smart on-line monitoring promotes optimal maintenance schedule

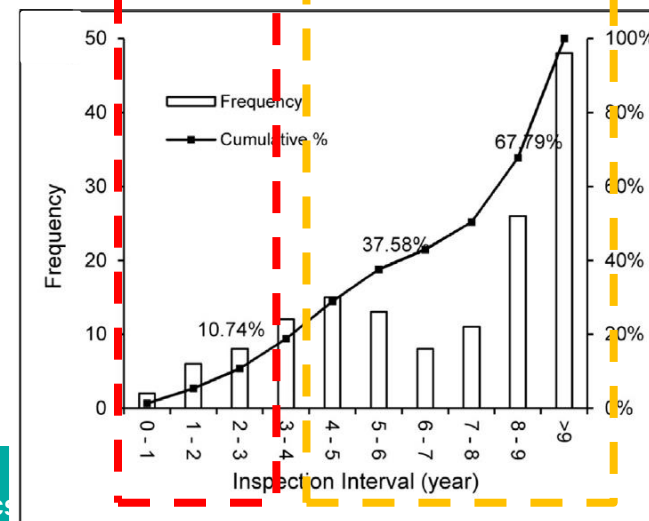
Objective I



Objective II

Implementing on-line monitoring system

However, in-line monitoring and periodic in-service inspections are necessary for these categories of equipment to find any unexpected changes in initial hypothesis, e.g. process conditions, corrosion rate, damage behavior, etc.



Smart Processing System

Feature No. 2: Smart on-line monitoring enables early warning for fault detection & diagnosis

Table 1: Comparison of perceived benefits between global and Malaysia organizations [6]

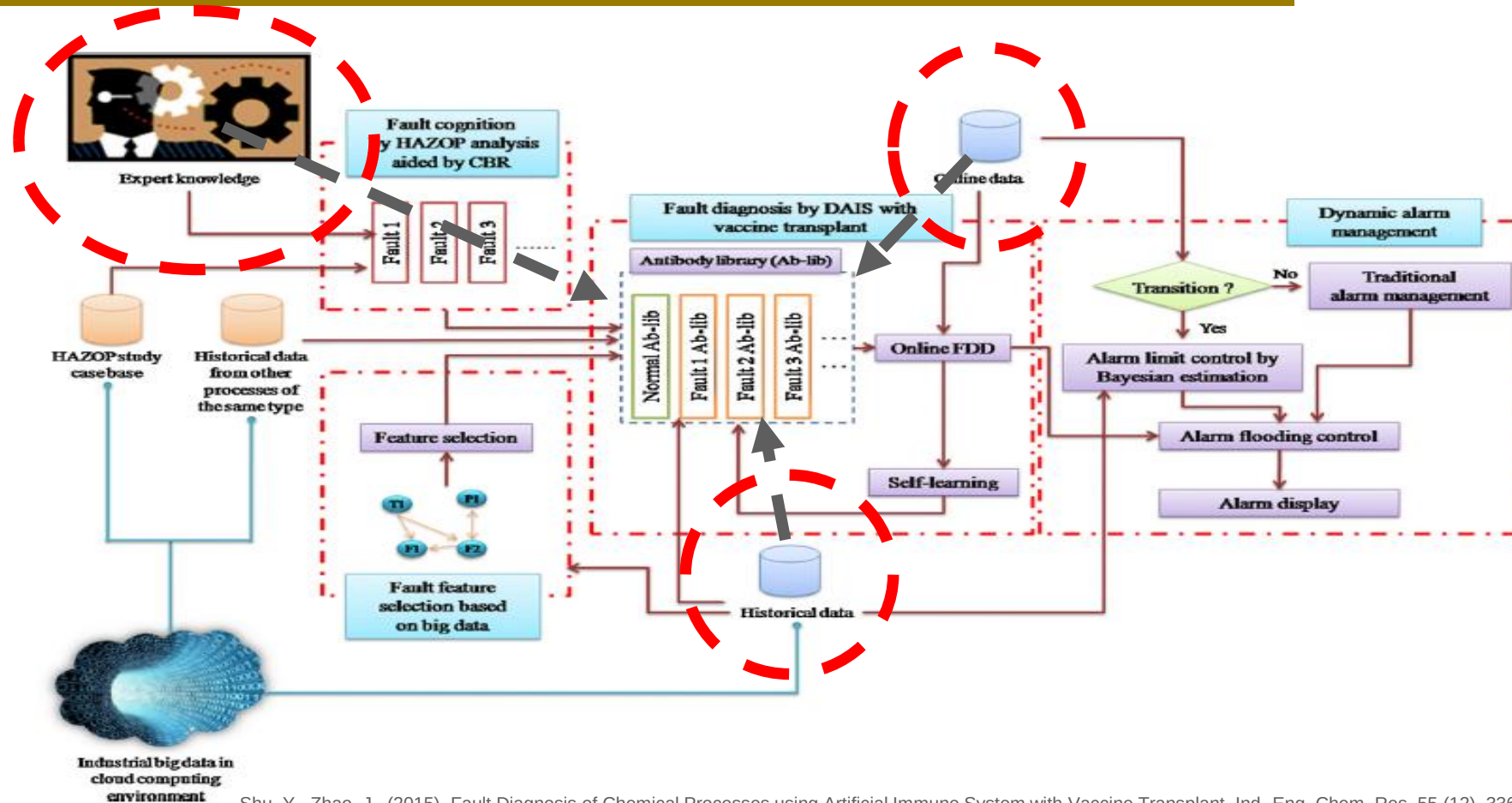
Benefit Group	Benefit	Global Organizations	Malaysian Organizations
Profitability	Decrease in number of failures occurrences	X	
	Reduction in direct inspection cost	X	X
	Fewer inspections	X	
Effective operations	Ensure equipment longevity	X	
	Increase in plant availability	X	X
	Elude unnecessary outages or system downtime	X	
	Minimum duration of shutdowns	X	
	Enhances understanding of integrity	X	
	Enables to make informed decision	X	
	Achieve operational excellence	X	X
	Increase effectiveness of inspection	X	X
Risk level	Reduction in the level of risk	X	X
	More precise assessment	X	
Inspection interval	Understanding of concepts	X	
	Tolerable inspection interval	X	X
	Attain an optimum schedule	X	
Societal Expectations	Fulfill societal expectations	X	X
Legal compliance	Maintain licensing permits	X	
	Avoid shutdown due to regulatory non-compliance	X	X


 Strengthening
 the operation
 integrity

] Mohamed, R., Hassan C.R.C., Hamid, M.D. (2017). "Implementing Risk-based Inspection Approach: Is it Beneficial for Pressure Equipment in Malaysia Industries?". Process Safety Progress, Vol. 00, No.00, 1-11.

Smart Processing System

Feature No. 3: Smart on-line monitoring enhances plant-wide control & failure prevention

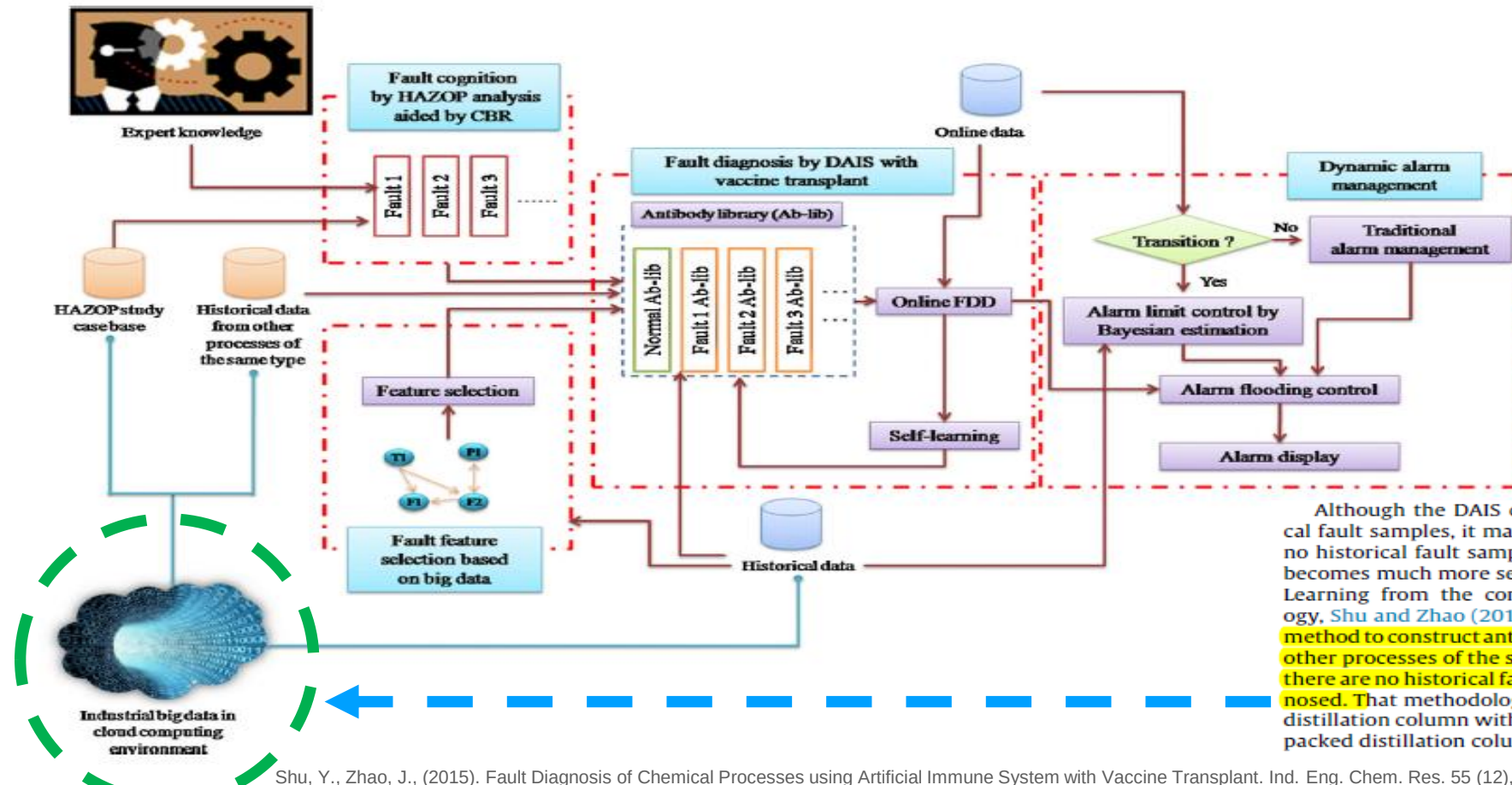


Reduce the likelihood of abnormal/catastrophic event

Shu, Y., Zhao, J., (2015). Fault Diagnosis of Chemical Processes using Artificial Immune System with Vaccine Transplant. Ind. Eng. Chem. Res. 55 (12), 3360-3371.

Smart Processing System

Feature No. 4: Smart on-line monitoring encourages smart data sharing through cloud networking



Resource sharing through cloud computing environment

Although the DAIS can work with a lower number of historical fault samples, it may still fail when the diagnosed process has no historical fault samples of a certain fault type. This issue even becomes much more severe in a new chemical facility or refinery. Learning from the concept of vaccine transplant in immunology, [Shu and Zhao \(2015\)](#) recently presented a vaccine transplant method to construct antibody libraries with fault data samples from other processes of the same type to make DAIS able to work when there are no historical fault samples in the target process to be diagnosed. That methodology was applied to fault diagnosis of a plate distillation column without fault data samples available by using a packed distillation column with fault data samples available.

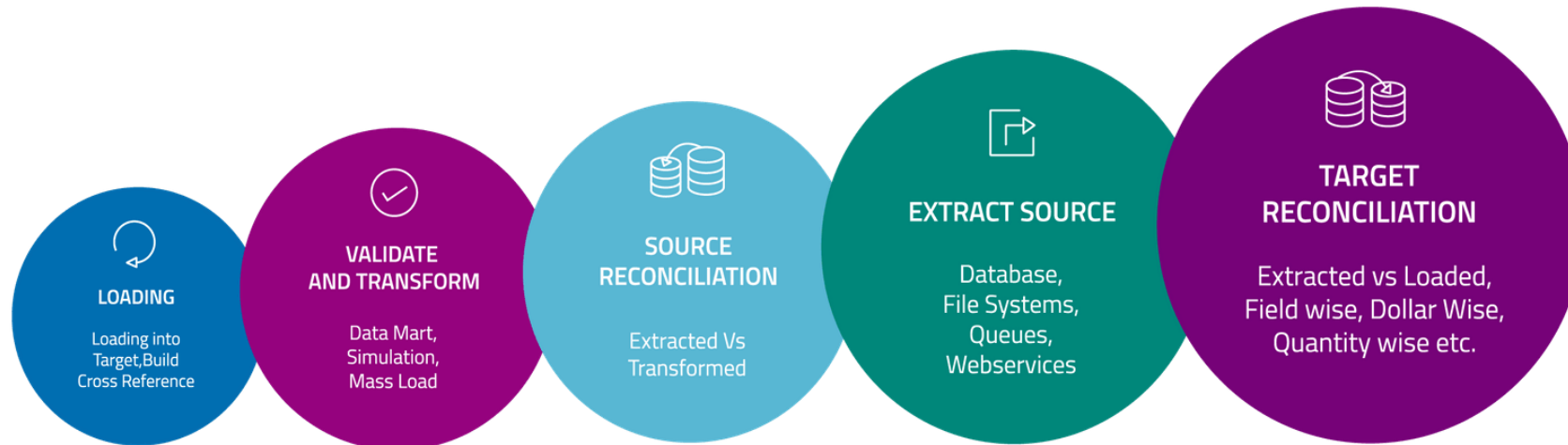
Shu, Y., Zhao, J., (2015). Fault Diagnosis of Chemical Processes using Artificial Immune System with Vaccine Transplant. Ind. Eng. Chem. Res. 55 (12), 3360-3371.

Exercise 2

Identify and explain the types of smart processing feature which currently implemented in your routine plant operation!

Smart Processing System

Data Reconciliation & Pre-treatment



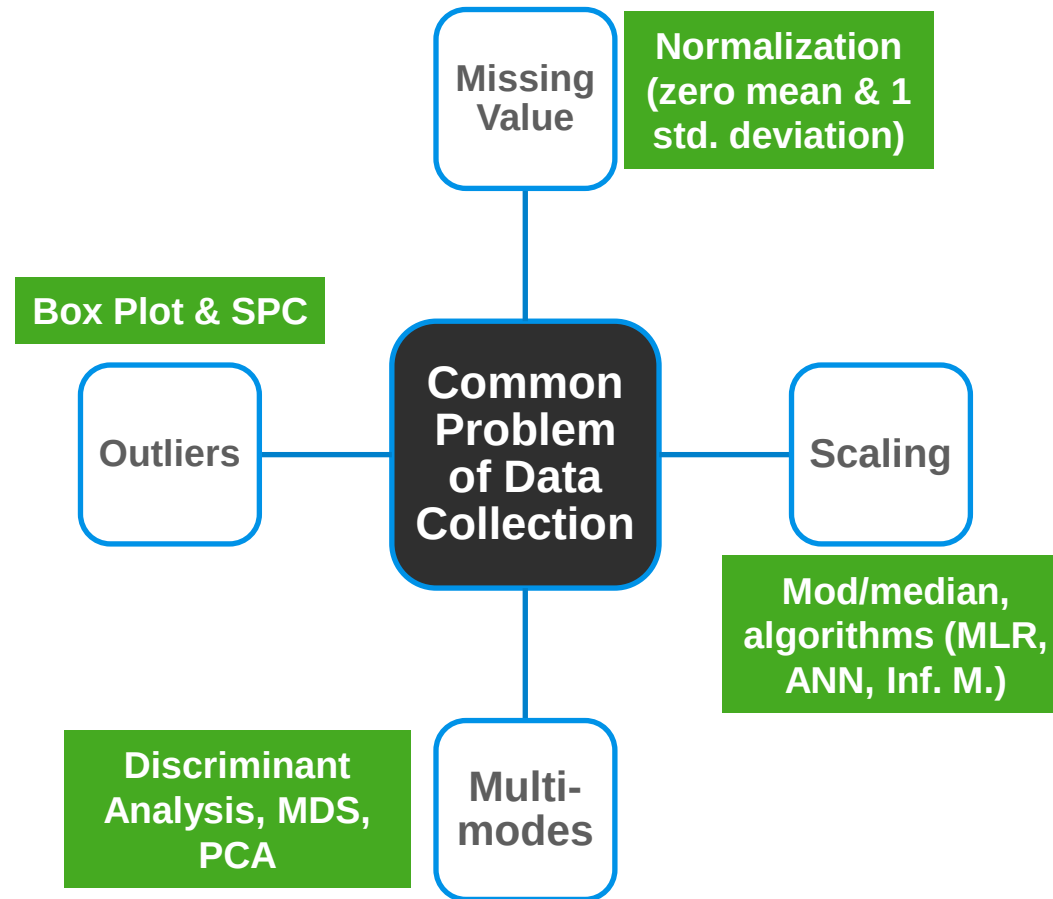
<https://www.chainsys.com/data-reconciliation>

Data reconciliation (DR) is a term typically used to describe a verification phase during a **data migration** where the target data is compared against original source data to ensure that the migration architecture has transferred the data correctly.

<https://www.experian.co.uk/business/glossary/data-reconciliation/>

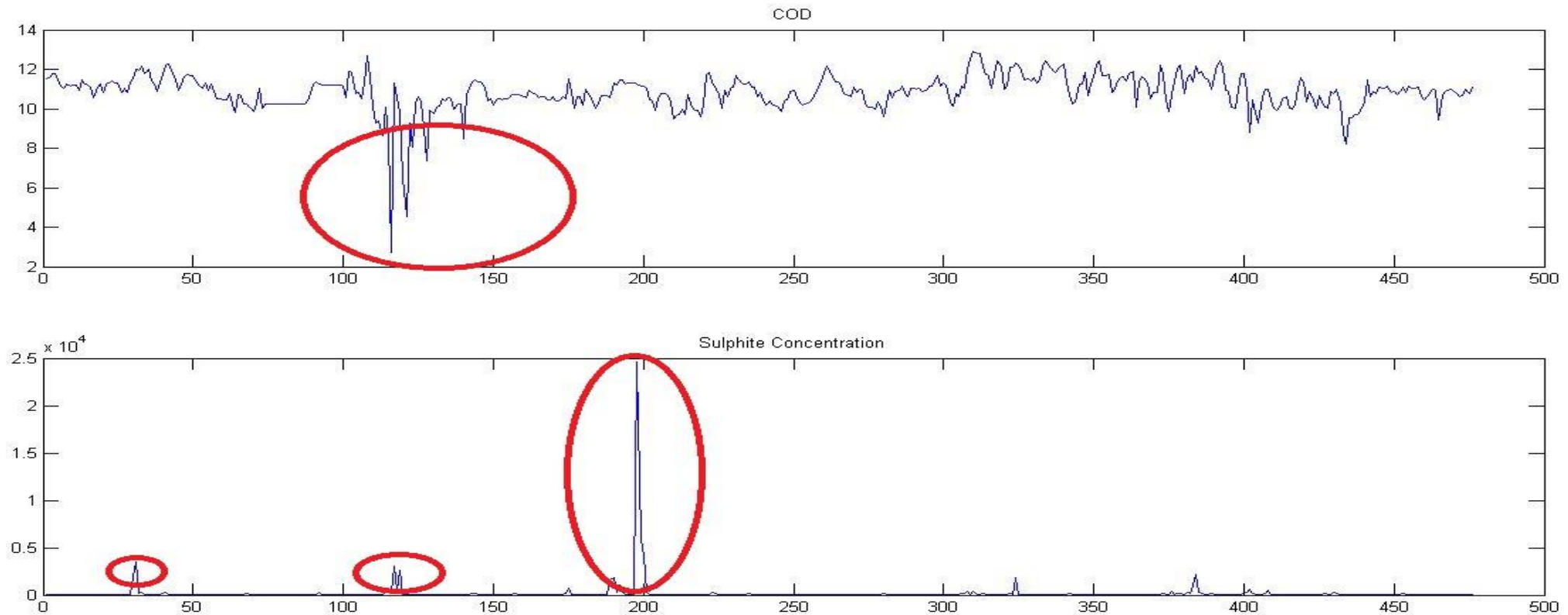
Smart Processing System

Data Reconciliation & Pre-treatment

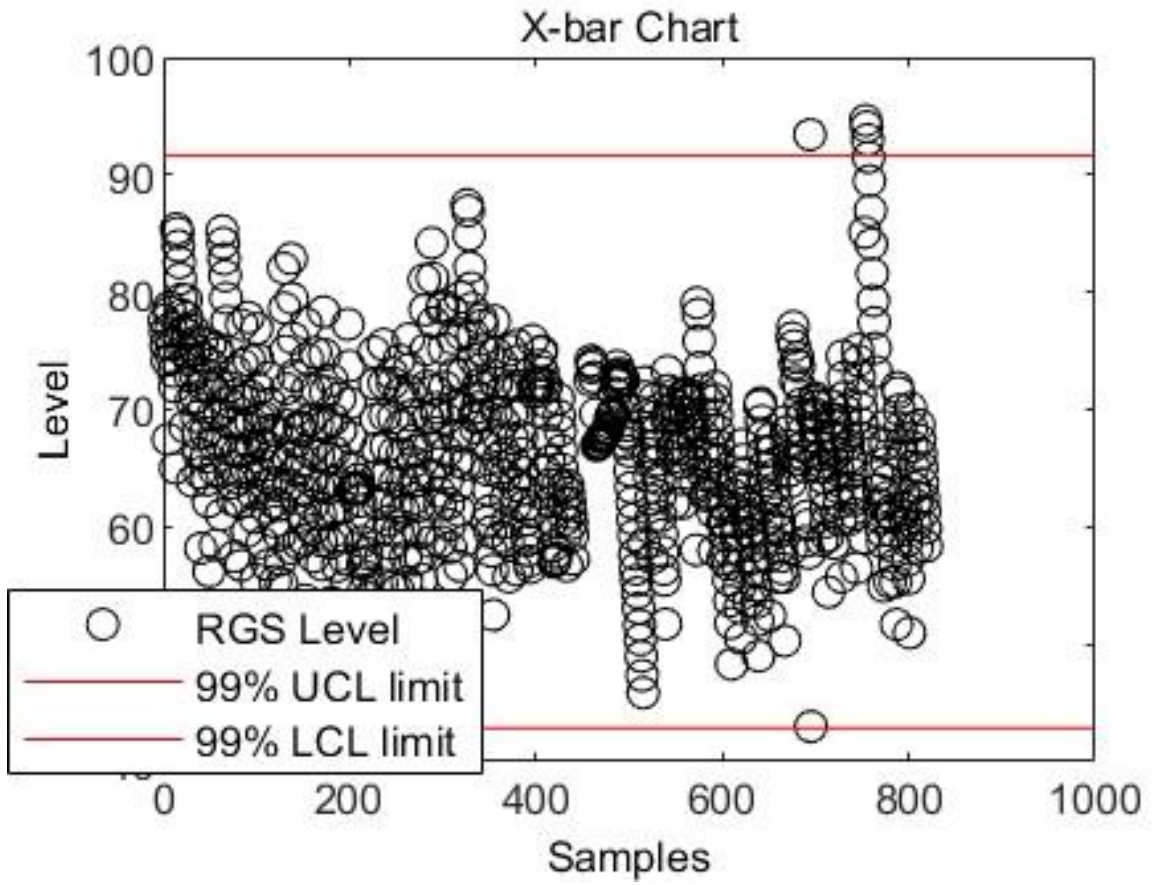
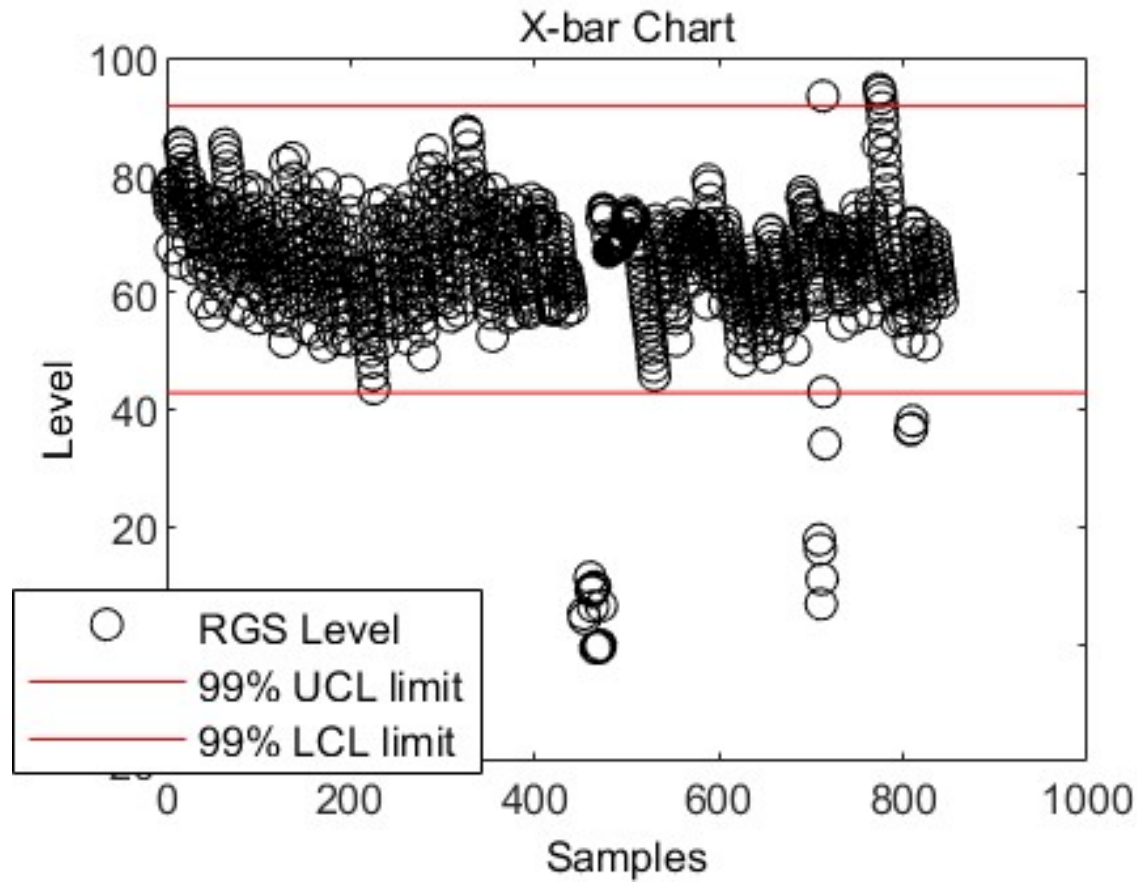


Smart Processing System

Outliers

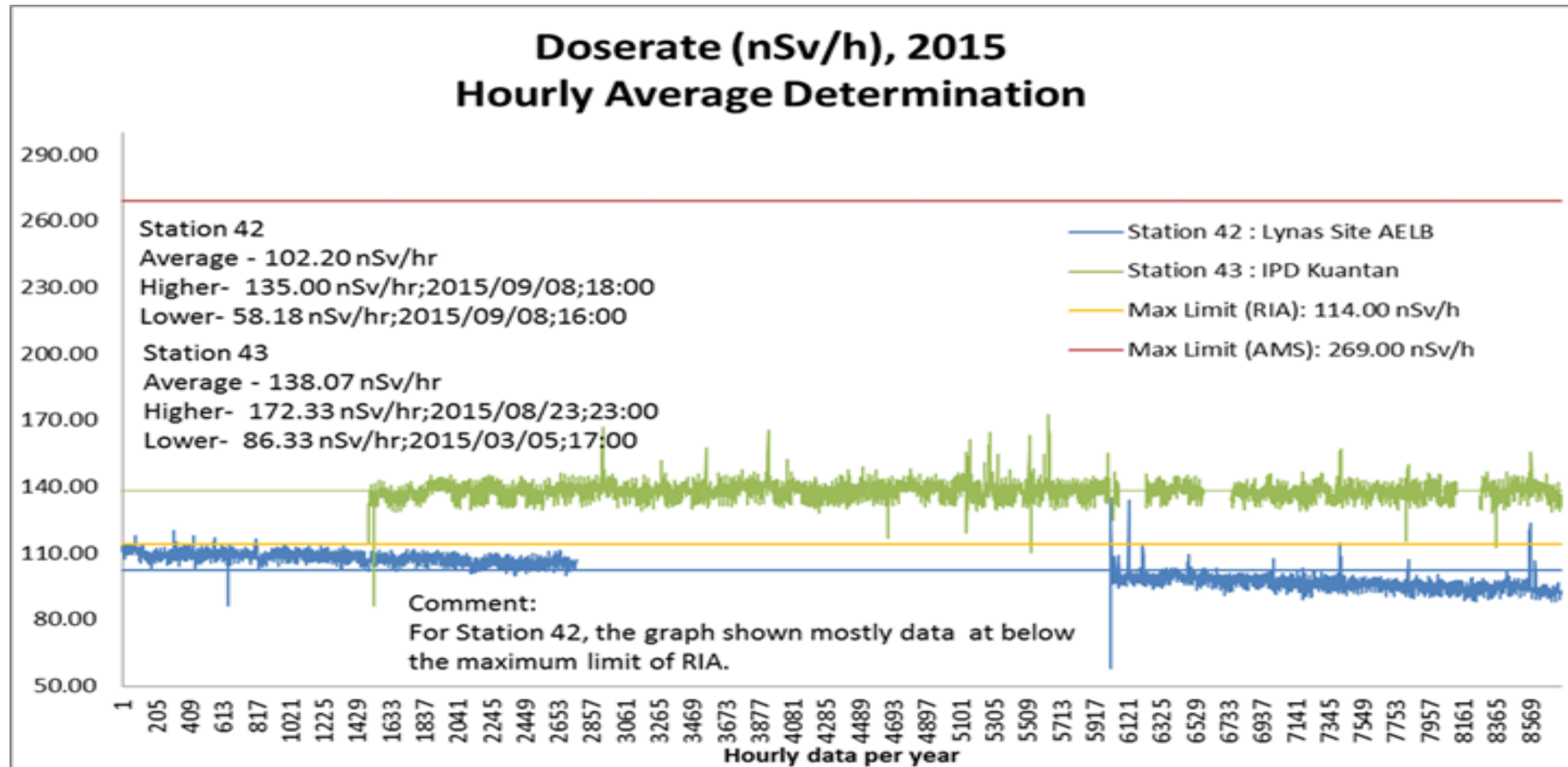


Smart Processing System Outliers



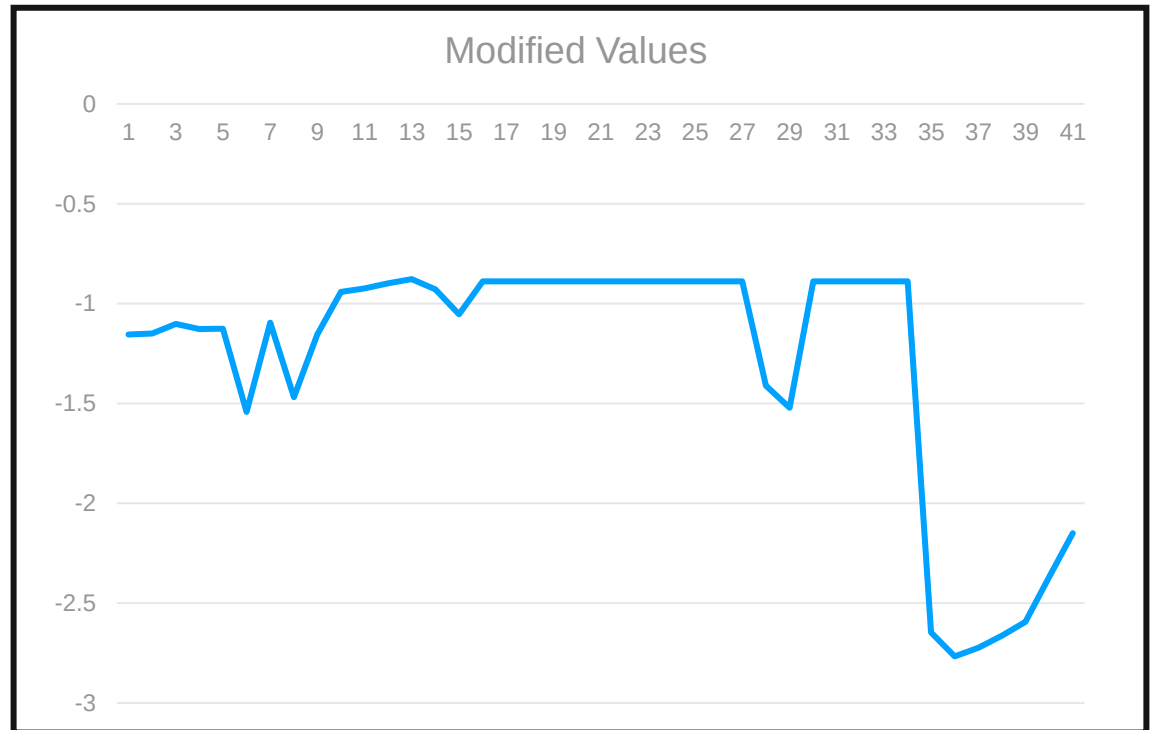
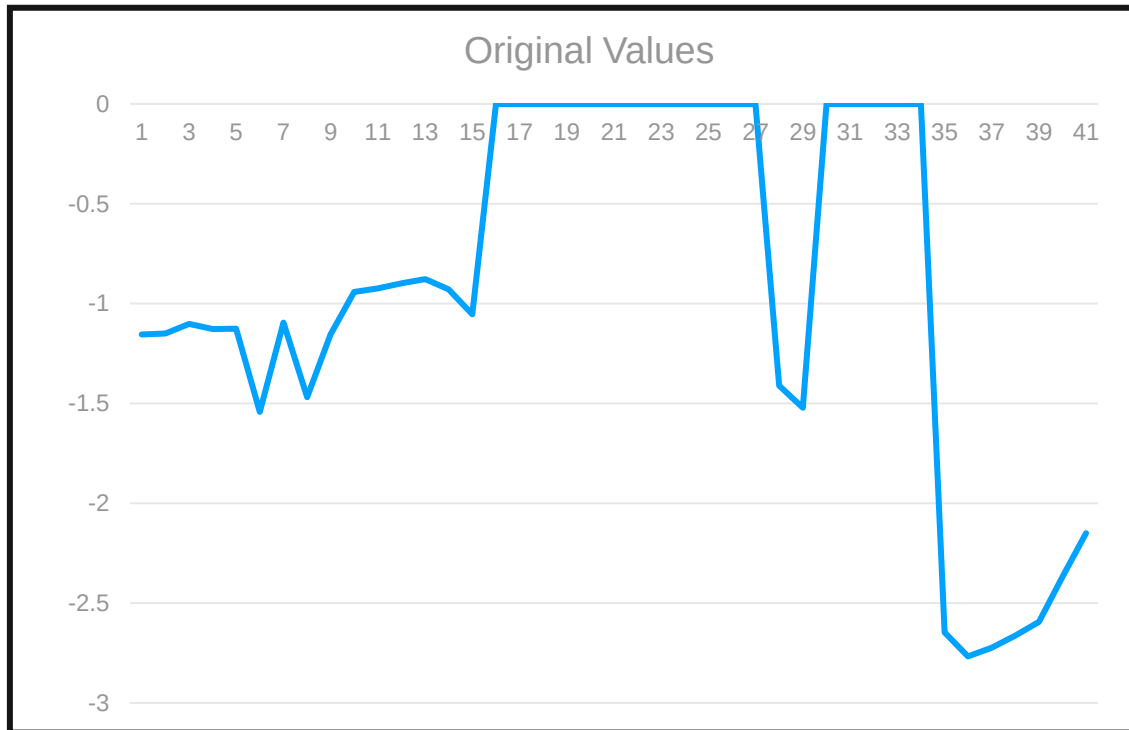
Smart Processing System

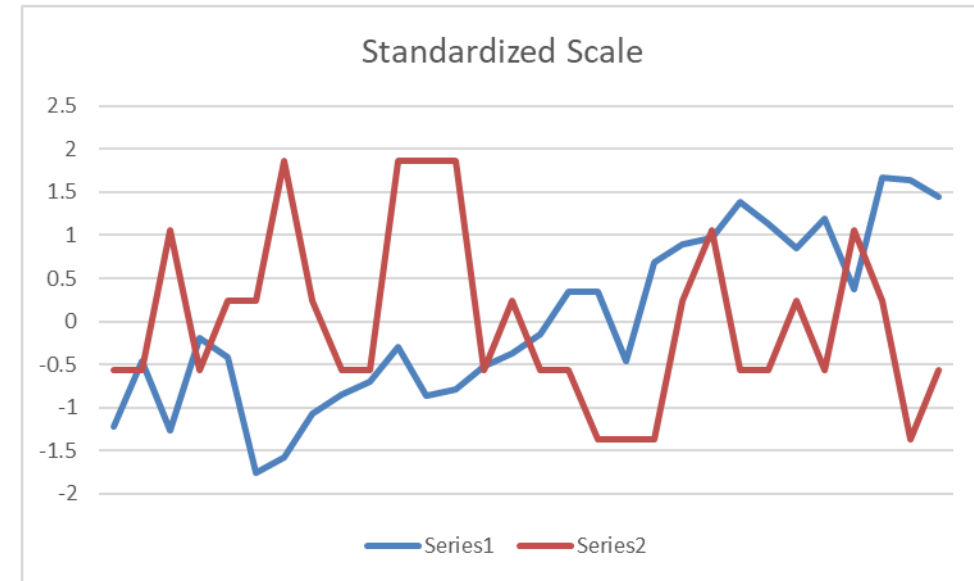
Missing Value



Smart Processing System

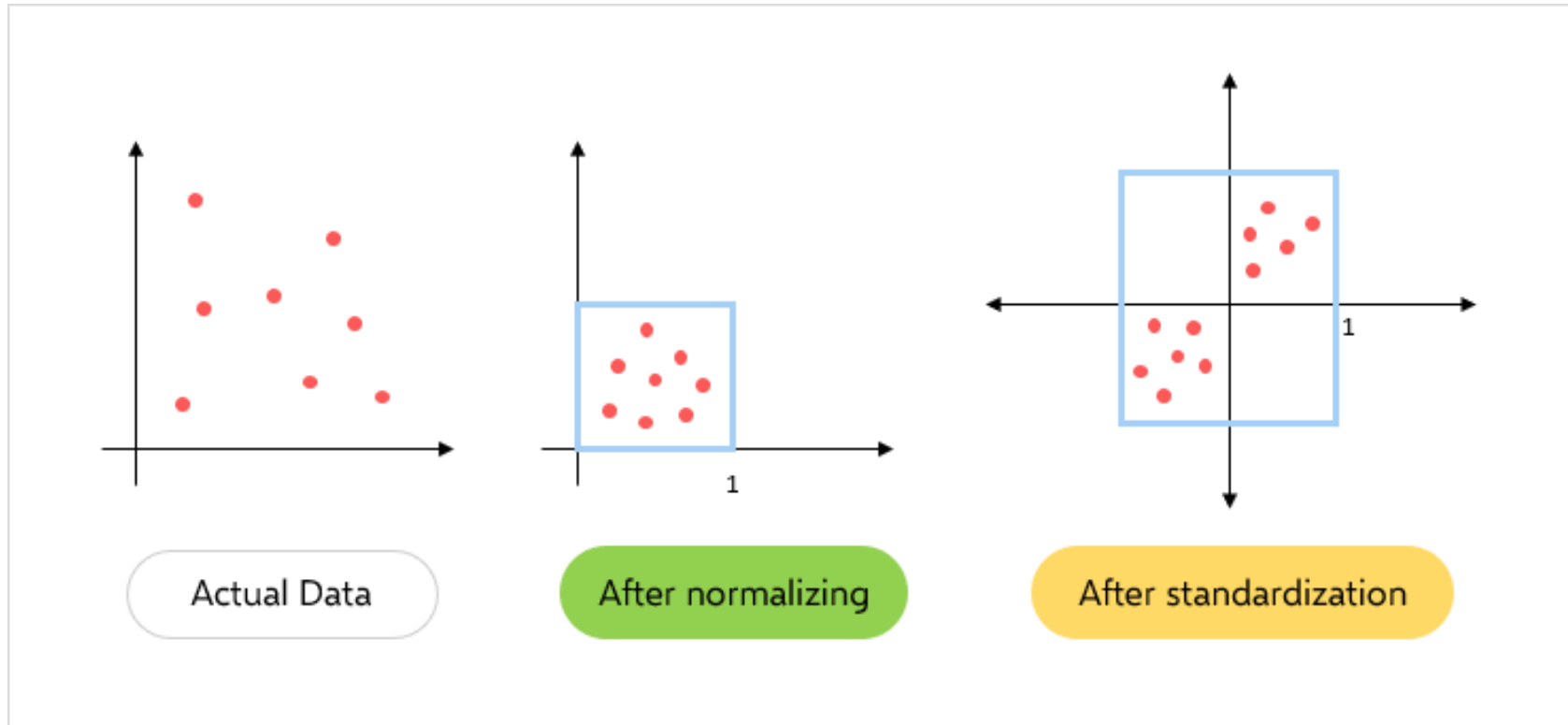
Missing Value





Smart Processing System

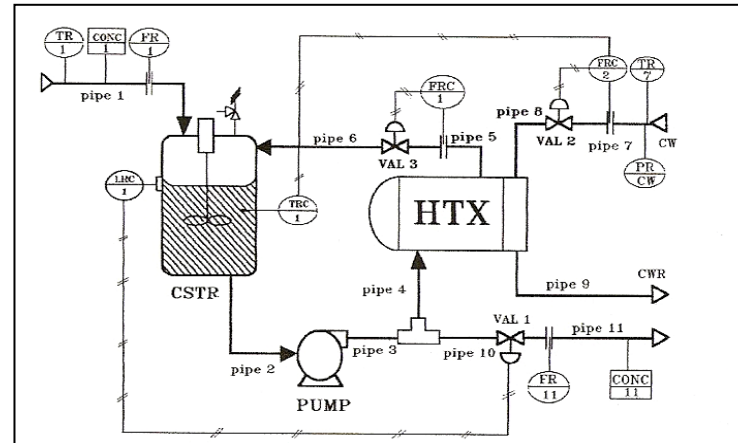
Scaling



<https://www.someka.net/blog/how-to-normalize-data-in-excel/>

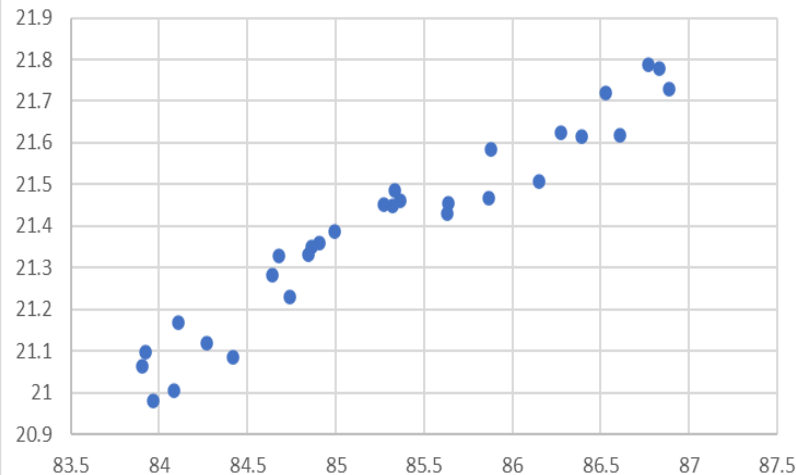
Smart Processing System

Multi-modes



Process			Instruments		
No.	Variables	Variable Names	No.	Variables	Variable Names
1	V1	Tank temperature	11	V11	Controller 1
2	V2	Tank level	12	V12	Controller 3
3	V3	Feed temperature	13	V13	Controller 2
4	V4	Inlet flow rate			
5	V5	Recycle flow rate			
6	V6	Outlet flow rate			
7	V7	Cooling water flow rate			
8	V8	Product concentration			
9	V9	Feed concentration			
10	V10	Heat exchanger entrance pressure			

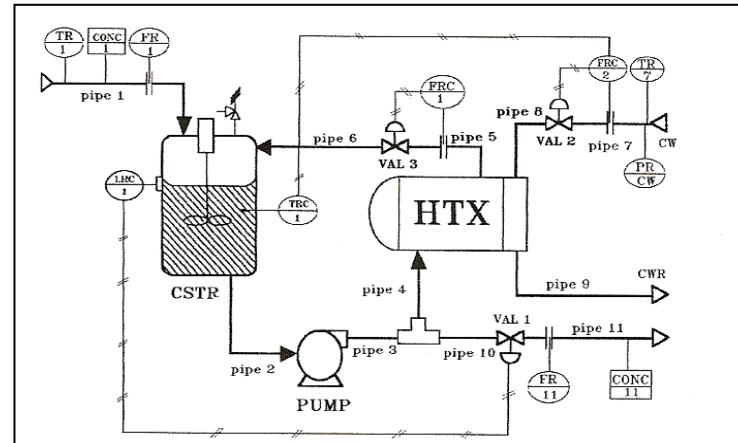
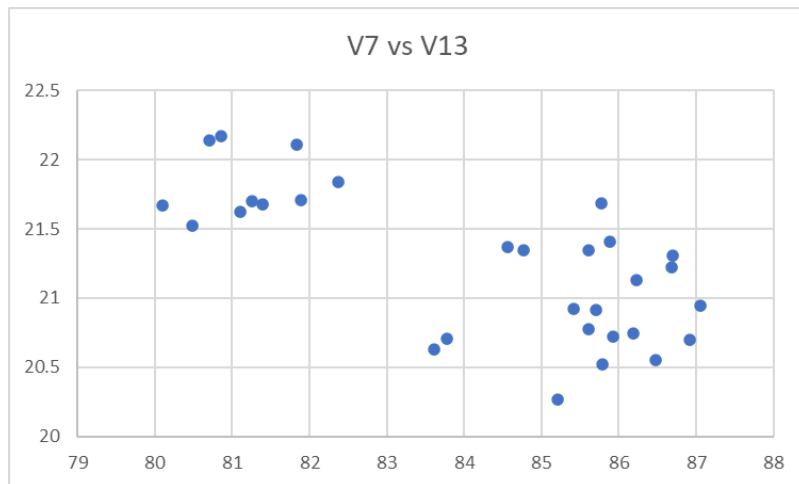
V7 vs V13



	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13
Column 1	1.000												
Column 2	-0.238	1.000											
Column 3	0.054	0.291	1.000										
Column 4	0.151	-0.050	-0.237	1.000									
Column 5	0.389	-0.152	0.053	-0.214	1.000								
Column 6	0.383	-0.236	0.146	0.176	0.293	1.000							
Column 7	-0.556	-0.036	-0.092	0.045	-0.532	-0.334	1.000						
Column 8	0.121	0.025	0.077	0.062	0.100	0.066	-0.189	1.000					
Column 9	-0.427	0.098	-0.204	-0.080	-0.013	-0.402	0.216	-0.070	1.000				
Column 10	0.078	0.284	0.185	0.010	0.063	0.315	-0.407	-0.093	-0.121	1.000			
Column 11	-0.395	0.207	-0.127	-0.160	-0.305	-0.995	0.355	-0.087	0.385	-0.302	1.000		
Column 12	0.391	-0.092	-0.016	-0.222	0.969	0.306	-0.500	0.064	0.015	0.121	-0.317	1.000	
Column 13	-0.594	-0.003	-0.173	-0.016	-0.497	-0.396	0.966	-0.200	0.241	-0.443	0.416	-0.473	1.000

Smart Processing System

Multi-modes

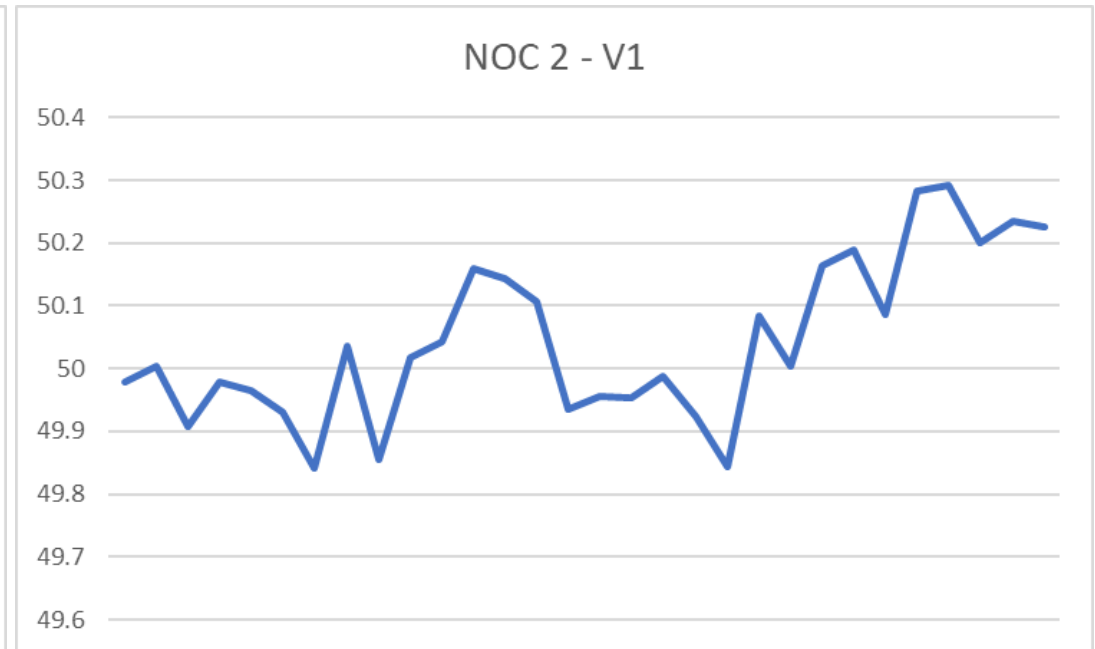
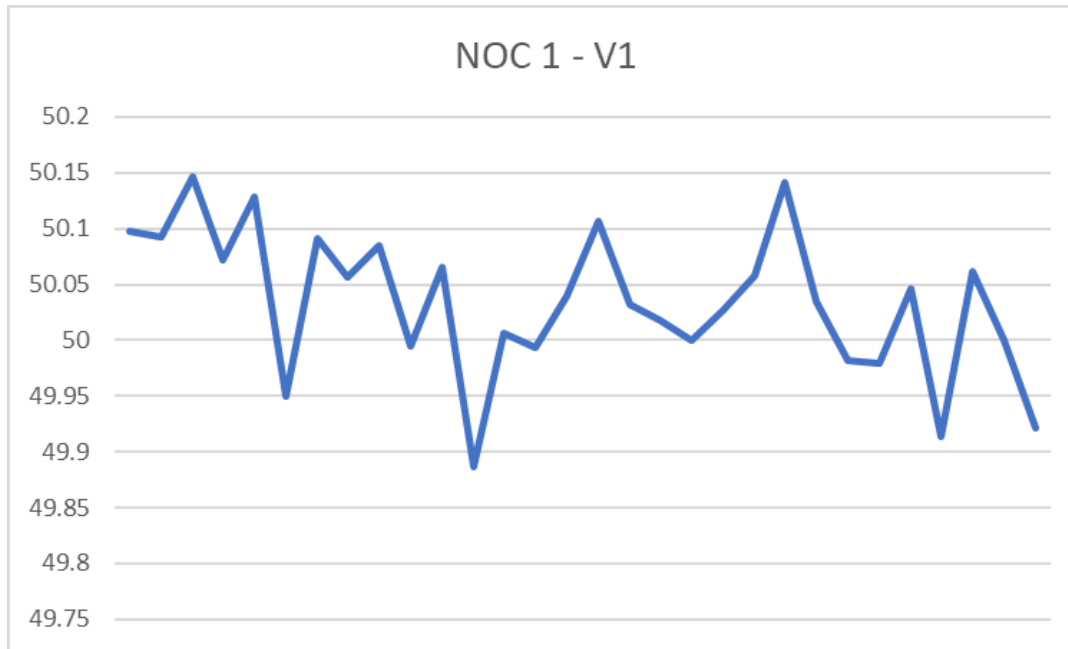


Process			Instruments		
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	Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7	Column 8	Column 9	Column 10	Column 11	Column 12	Column 13
Column 1	1.000												
Column 2	-0.182	1.000											
Column 3	0.202	0.114	1.000										
Column 4	0.110	-0.204	-0.113	1.000									
Column 5	0.009	-0.377	0.306	0.191	1.000								
Column 6	0.146	0.154	0.042	0.178	0.128	1.000							
Column 7	-0.690	0.007	-0.088	-0.077	0.118	-0.130	1.000						
Column 8	-0.445	-0.011	0.041	-0.170	0.129	0.098	0.260	1.000					
Column 9	0.358	-0.267	-0.116	0.104	0.122	0.037	-0.184	-0.296	1.000				
Column 10	-0.511	0.371	0.112	-0.008	0.028	0.235	0.340	0.212	-0.286	1.000			
Column 11	-0.169	-0.121	-0.204	-0.268	-0.149	-0.896	0.106	-0.068	-0.084	-0.201	1.000		
Column 12	-0.053	-0.420	0.274	0.147	0.653	0.196	0.269	0.044	0.276	0.033	-0.223	1.000	
Column 13	0.592	-0.083	0.018	0.214	0.023	-0.196	-0.700	-0.187	0.173	-0.457	0.187	-0.163	1.000

Smart Processing System

Multi-modes



Exercise 3

Detect any of data reconciliation issues on the operation data that collected!